Francesco Deangelis

Curriculum vitae

Münster, NRW, Germany
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francescodeangelis.github.io

Experience

10/2024–present **Postdoctoral Researcher in Mathematics**, *University of Münster*, Germany
Mentor: Prof. Caterina Ida Zeppieri

10/2023–11/2023 **Short-Term Visitor**, *Institute for Advanced Study*, Princeton, NJ, USA
Visiting Prof. Camillo De Lellis (1 October – 30 November 2023)

Education

11/2020–7/2024 **PhD in Mathematics**, *Gran Sasso Science Institute*, L'Aquila, Italy
cum laude
Advisor: Prof. Camillo De Lellis (IAS, Princeton)
Thesis: Boundary regularity for the Mumford–Shah functional with Dirichlet boundary conditions

10/2017–3/2020 **MSc in Pure and Applied Mathematics**, *University of Rome Tor Vergata*, Italy
110/110 cum laude
Advisor: Prof. Michiel Bertsch
Thesis: A free boundary problem arising in the theory of binary fluids
Award: €2000 prize for the master's thesis

10/2014–7/2017 **BEng in Mechanical Engineering**, *University of Rome Tor Vergata*, Italy
110/110 cum laude
Advisor: Prof. Giovanni Bellettini
Thesis: A variational approach to fracture evolution

2013–2014 **BSc in Economics and Industrial Organization**, *University of Warwick*, UK
Left before completion in April 2014 to pursue studies more aligned with my academic interests

2008–2013 **Diploma di Maturità Classica**, *Liceo Classico Ugo Foscolo*, Albano Laziale (Rome), Italy
100/100 cum laude

Publications

6) Francesco Deangelis, Camillo De Lellis, and Matteo Focardi. A remark on the global minimality of the cracktip, 2026. `arXiv:2610.33900`.

5) Francesco Deangelis. Classification of boundary blow-ups for the Mumford-Shah functional with Dirichlet boundary conditions, 2026. `arXiv:2610.23513`.

4) Francesco Deangelis. Solution of the Mumford–Shah conjecture, 2026. `arXiv:2609.26732`.

3) Francesco Deangelis, Matteo Focardi, and Caterina Ida Zeppieri. Stochastic homogenization of fractional obstacle problems, 2026. `arXiv:2604.09896`.

2) Francesco Deangelis. Boundary regularity for the Mumford-Shah functional with Dirichlet boundary conditions, 2024. arXiv:2404.07162.

1) Roberto Benzi, Michiel Bertsch, and Francesco Deangelis. A free boundary problem for binary fluids. *Interfaces Free Bound.*, 23(4):485–506, 2021. URL: https://ems.press/journals/ifb/articles/3324283, doi:10.4171/IFB/462.

Talks

01/07/2026 **Stochastic homogenization of fractional obstacle problems**, *Oberseminar Stochastik, University of Münster*, Germany

11/02/2026 **Boundary regularity for the Mumford–Shah functional with Dirichlet boundary conditions**, *XXXV Convegno Nazionale di Calcolo delle Variazioni*, Riccione, Italy

30/01/2026 **Stochastic homogenization of fractional obstacle problems**, *Analysis Seminar, University of Florence*, Italy

03/12/2025 **Boundary regularity for the Mumford–Shah functional with Dirichlet boundary conditions**, *Analysis Seminar, University of Parma*, Italy

26/11/2025 **Stochastic homogenization of fractional obstacle problems**, *CASA Colloquium, Eindhoven University of Technology*, Netherlands

29/10/2025 **An introduction to stochastic homogenization**, *University of Münster*, Germany

10/04/2025 **Homogenization on randomly perforated domains**, *Postdoc Colloquium, University of Münster*, Germany

03/07/2024 **Boundary regularity for the Mumford–Shah functional with Dirichlet boundary conditions**, *Analysis Seminar, Gran Sasso Science Institute*, L'Aquila, Italy

Teaching

2025 **Exercise Class Instructor - Applied Functional Analysis**, *University of Münster*

2018-2019 **Tutor - Mathematical Analysis 1**, *University of Rome Tor Vergata*

Languages

Italian Native

English Fluent — TOEFL iBT 115/120

Computer skills

Web programming PHP, MySQL, AJAX, CodeIgniter, jQuery, JavaScript, HTML, CSS, Bootstrap, Python, Django, Jekyll

Mobile programming Flutter

Scientific programming MATLAB, Python, C